

TaxaAssign

Bayesian taxonomic assignment from match scores and priors. Part of the TaxaID ecosystem.

Overview

TaxaAssign makes a consensus taxonomic assignment for the several hypothesized matches to each observation. It uses Bayes' theorem to multiply the likelihood that a match score corresponds to a particular species (from TaxaLikely) by the prior probability that the species would be selected at random from a sample of similar species at that site and habitat (from TaxaExpect). After normalization and Monte Carlo simulation, each candidate receives a posterior probability. If a single candidate has strong support, it is assigned as the consensus taxon; otherwise, a coarser rank (genus, family, etc.) is assigned via lowest common ancestor. All competing hypotheses and their probabilities are retained in the output. The user can also update priors iteratively: if one observation strongly supports species A, that evidence can sharpen the prior for species A in other observations from the same sample.

Two workflows:

- **Full Bayesian** – TaxaLikely's trained likelihood model + TaxaExpect's occurrence-modelled priors. The recommended pathway for any real analysis: publication-quality, and every one of this ecosystem's real production workflows uses it exclusively.
- **LLM-shortcut** – an approximate *stand-in* for the pathway above, substituting an exponential score-weighting proxy for TaxaLikely's modeled likelihood and an LLM's biogeographic judgment for TaxaExpect's modeled occurrence prior. It exists for exploratory analysis when a trained TaxaLikely model or TaxaExpect priors aren't available yet (or as a quick comparison against the Full Bayesian result on data you already have both for). This fast path is better than relying on scores, but **not** the statistically defensible outcome that most users want.

Installation

```
# Requires TaxaTools (foundation package)
devtools::install("path/to/TaxaTools")
devtools::install("path/to/TaxaAssign")
```

Quick Start

Full Bayesian workflow (recommended)

```
out <- run_bayesian_pipeline(
  match_df      = match_obj,          # from TaxaMatch
  model_params  = trained_model,      # from TaxaLikely
  taxaexpect_priors = priors,         # from TaxaExpect
  site = list(grid_id = "Grid_34p1_m119p1", main_habitat = "Marine")
)
head(out$consensus)
```

LLM-shortcut workflow (fast approximation – no trained model/priors needed)

```
library(TaxaAssign)

out <- run_llm_pipeline(
  match_df      = match_obj,      # from TaxaMatch
  geographic_hint = "Southern California",
  barcode_term   = "12S"
)
head(out$consensus)
```

Step-by-step

```
# 1. Join priors to likelihood output
joined <- join_priors(likelihoods, priors,
  site = list(grid_id = "G1", main_habitat = "Marine"))

# 2. Compute posteriors via Monte Carlo
posteriors <- compute_posterior(joined, n_sims = 1000)

# 3. Consensus taxonomy (LCA among plausible hypotheses)
consensus <- posterior_consensus(posteriors)

# 4. Generate report
report <- generate_report(posteriors, consensus)
```

Key Functions

Core assignment: - `compute_posterior()` – Bayes’ theorem with MC uncertainty - `assign_taxa_llm()` – LLM-shortcut (priors + likelihoods in one call; accepts named unreferenced species from `TaxaLikely::suggest_unreferenced` via `unreferenced_taxa` – see `TaxaLikely`’s README for the full unreferenced-species mechanism) - `join_priors()` – merge `TaxaExpect` priors with likelihood output

Consensus: - `posterior_consensus()` – LCA from posterior probabilities - `score_consensus()` – conventional score-based consensus - `update_prior_from_consensus()` – empirical Bayes refinement

High-level wrappers: - `run_bayesian_pipeline()` – Full Bayesian (`TaxaLikely` + `TaxaExpect` -> posteriors) – recommended - `run_llm_pipeline()` – LLM-shortcut approximation (`match_df` -> posteriors in one call, no trained `TaxaLikely` model or `TaxaExpect` priors needed)

Context and reporting: - `build_context()` – auto-populate site context from taxon names - `generate_report()` – publication-ready Methods + Results text - `report_assign()` – lightweight section for `assemble_report()`

Statistical Methods

`TaxaAssign` computes Bayesian posterior probabilities for each candidate taxonomic assignment:

$$P(H_i | D) = \frac{L(D | H_i) \times \pi(H_i)}{\sum_j L(D | H_j) \times \pi(H_j)}$$

where L is the likelihood (TaxaLikely’s trained model, or the LLM-shortcut pathway’s exponential score-weighting proxy) and π is the prior (TaxaExpect’s occurrence model, or the LLM-shortcut pathway’s LLM-estimated biogeographic prior).

- **Monte Carlo uncertainty propagation:** priors are modelled as $\text{Beta}(\alpha, \beta)$ distributions and likelihoods as $\text{Normal}(\text{mean}, \text{sd})$; 1000 simulations (default) propagate both sources of uncertainty into posterior means, SDs, and confidence scores (fraction of simulations won)
- **Two workflows:** the full Bayesian workflow uses calibrated likelihoods from TaxaLikely’s hierarchical model and spatially-explicit priors from TaxaExpect’s kernel-based composition estimator; the LLM-shortcut workflow uses exponential score weighting ($L_i = e^{\lambda s_i}$) and LLM-estimated priors with information-quality-driven Beta concentration
- **Posterior consensus:** hypotheses below `min_posterior` (default 0.05) are dropped, then the remainder are accumulated in decreasing order until they reach `cumulative_threshold` (default 0.90) of the named-taxon posterior mass; if multiple taxa remain, the lowest common ancestor (LCA) determines the consensus rank; optional downranking (`species_reference`, off by default) refines coarse assignments when only one finer-rank taxon exists at the study site
- **Empirical Bayes refinement:** species confidently identified in one observation receive boosted priors in unresolved observations from the same study, analogous to shrinkage estimators (Efron and Morris 1973)
- **Dark diversity fallback:** “dark diversity” is an ecological concept (Pärtel et al. 2011) for species that belong to the regional species pool and could plausibly occur at a site given its environmental conditions, but have not actually been observed there. TaxaExpect computes per-species Tier 3 (“undetected species”) estimates using this concept – see TaxaExpect’s README for the full occurrence-modeling mechanism. For species with no prior row from TaxaExpect at all, `join_priors()` builds its own fallback from those Tier 3 estimates – site-level or global averaging, or (with `singleton_taxonomy`) mass-conserving hierarchical group priors – preventing false negatives from incomplete occurrence data. See `join_priors()`’s own documentation for the full mechanism.

For the full statistical derivation, assumptions, and references, see `inst/TaxaAssign_supplemental_methods.md`.

Vignettes

- Taxonomic Assignment – full workflow
- TaxaID Ecosystem Overview – cross-package guide

Part of TaxaID

TaxaAssign is the convergence point of the TaxaID ecosystem. It receives likelihoods from TaxaLikely and priors from TaxaExpect, then passes assignments to TaxaFlag for quality screening.

Ecosystem: TaxaLikely + TaxaExpect -> **TaxaAssign** -> TaxaFlag

See the TaxaID README for ecosystem overview and installation instructions.

Citation

Lafferty, K.D., 2026, TaxaID – A modular R ecosystem for Bayesian taxonomic assignment: U.S. Geological Survey software release, <https://doi.org/10.5066/xxxxxx>.

Software Requirements

- R ($\geq 4.1.0$; R Core Team 2025)
- TaxaTools (foundation package)
- TaxaLikely and TaxaExpect (for the Bayesian workflow; in Suggests)
- An Application Programming Interface (API) key for an LLM provider (Anthropic Claude, Google Gemini, OpenAI, or local Ollama – see TaxaTools) is needed for the LLM-shortcut workflow

All dependencies are declared in the DESCRIPTION file and installed automatically.

Developed with Claude Code (Anthropic PBC, San Francisco, California).

References

- Efron, B. and Morris, C. (1973). Stein’s estimation rule and its competitors – an empirical Bayes approach. *Journal of the American Statistical Association*, 68(341), 117–130.
- Pärtel, M., Szava-Kovats, R. and Zobel, M. (2011). Dark diversity: shedding light on absent species. *Trends in Ecology & Evolution*, 26(3), 124–128.
- R Core Team (2025). R: A Language and Environment for Statistical Computing. V.4.5.2. R Foundation for Statistical Computing, Vienna, Austria. <https://www.r-project.org>